

Combining Overlap Correction and Back-to-back Regression Methods to Analyse Eye-tracking-EEG Co-registration Data during Natural Reading.

<Msc> Thesis Midterm Talk - <Yanhong Xu>

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Introduction and related works

Natural reading

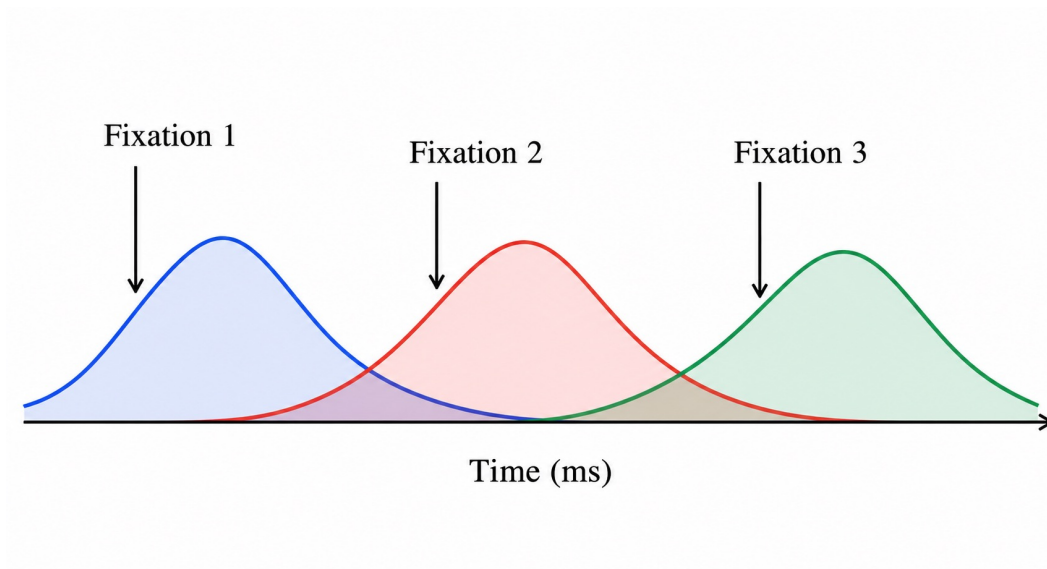
The Marketer: Alchemist, Magician, Sorcerer and Medicine Man

Is marketing an art or a science? Perhaps marketing is more like sorcery. Think of a sorcerer collecting ingredients from different sources and mixing them into a potion, accompanied with the magical effect of a flash of light and the illusion to follow. To some extent this fits with Culler's vision of a marketer as a 'mixer of ingredients'. Of course sorcerers are more mythical than real, but if we stay with this myth it may help to dispel some of the myths surrounding 'marketing'.

Eye movement · visual processing · lexical processing ·
contextual integration · comprehension · next eye-movement decision

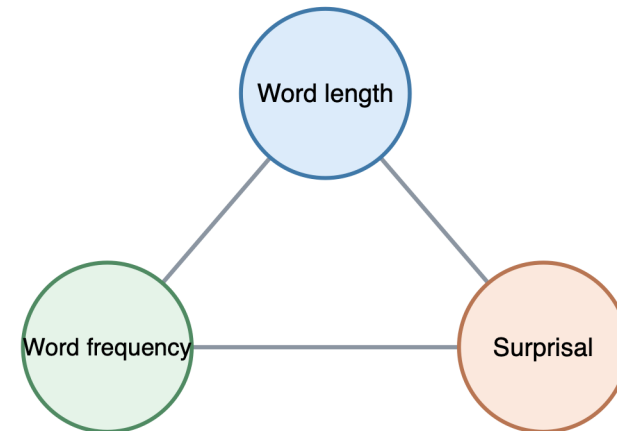
Natural reading adds two challenges

1. Temporal Overlap



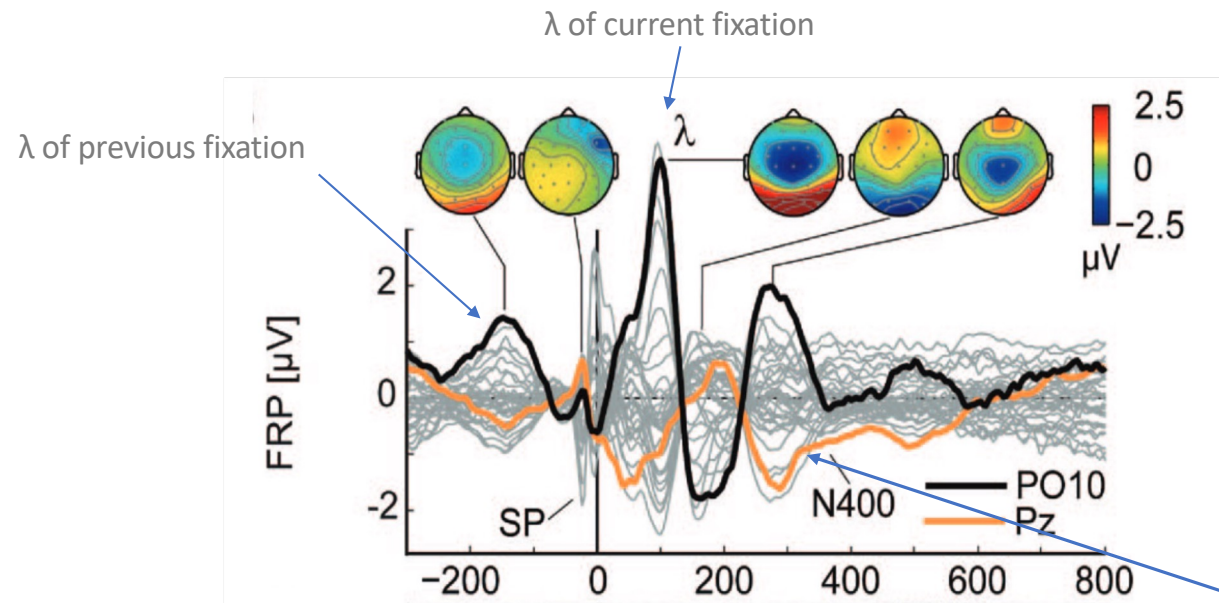
Which fixation generated this EEG activity?

2. Predictor Collinearity



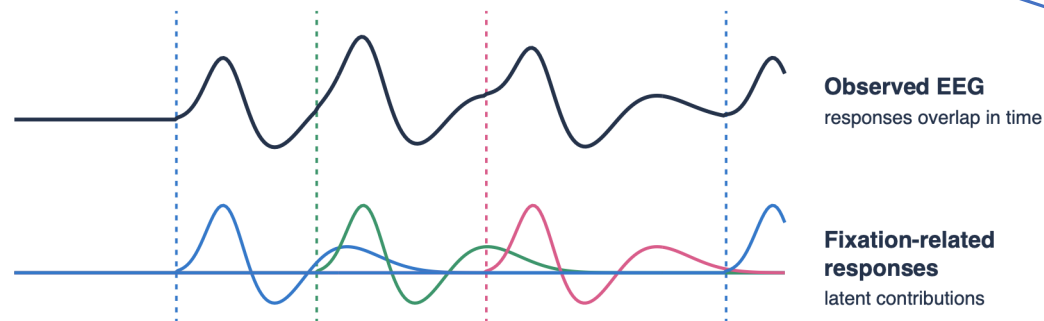
Which linguistic feature explains this EEG activity?

Linear deconvolution (rERP) for overlap correction

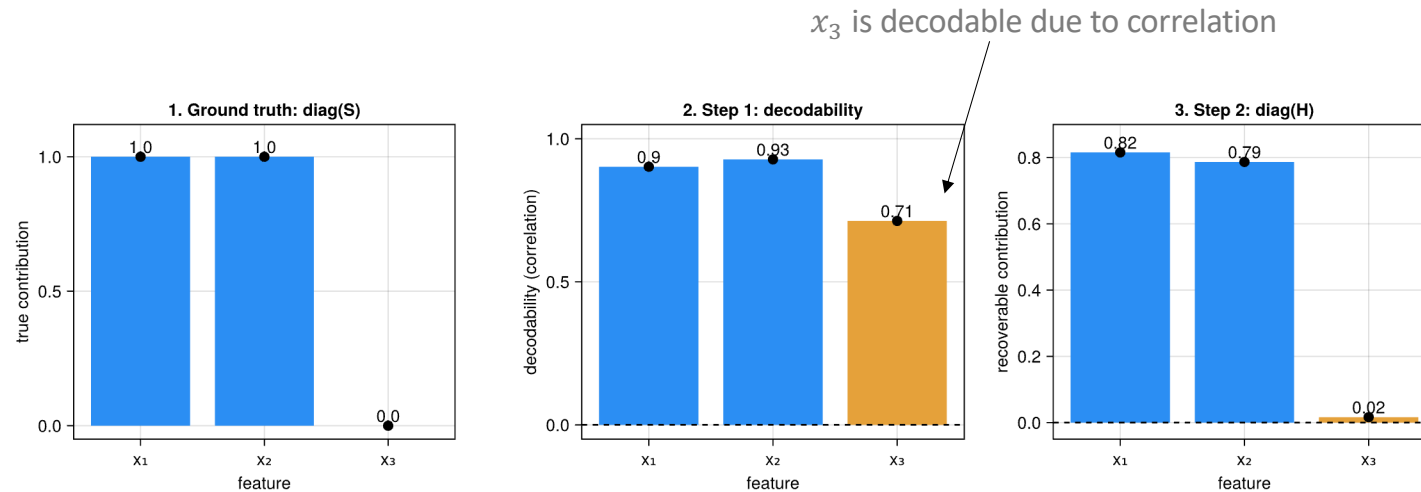


- λ : early visual component.
- N400: lexico-semantic response component.

N400 of fixation n overlaps with λ from fixation n+1



Back-to-Back Regression disentangles correlated predictors



B2B estimates: Does the EEG contain information specifically about this predictor?

Analysis methods for combining rERP and B2B

		Collinearity	
		Low	High
Overlap	None	Standard decoding <i>baseline</i>	Back-to-back regression (B2B) <i>disentangles collinear predictors</i>
	High	Overlap-corrected decoding <i>deconvolution rERP</i>	One-step: deconvolution rERP + B2B Two-step: GLM-reconstructed trials → B2B

Evaluate in two settings:

1. Controlled simulations

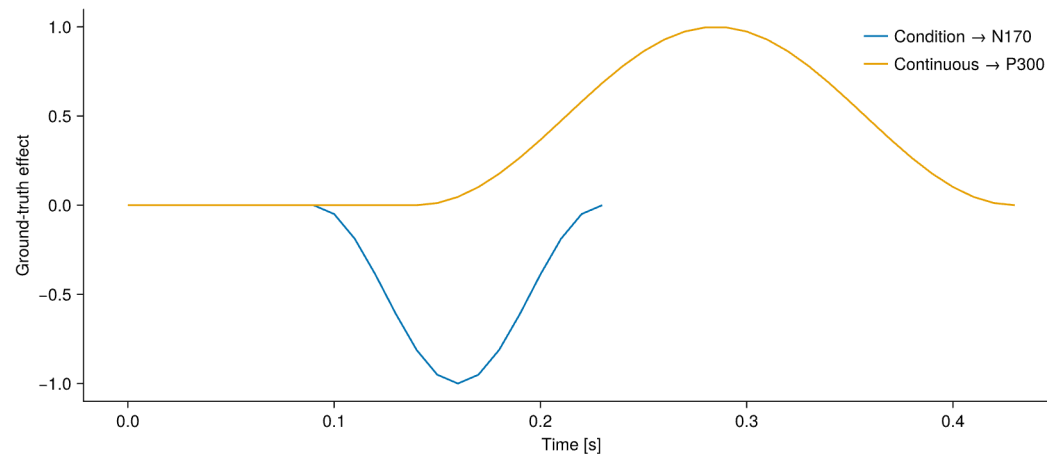
2. ROAMM natural reading

Simulations

- Main simulation: Condition + Continuous
- Additional simulation: Correlated continuous

Main simulation: condition + continuous predictors

Ground truth:



Four cases:

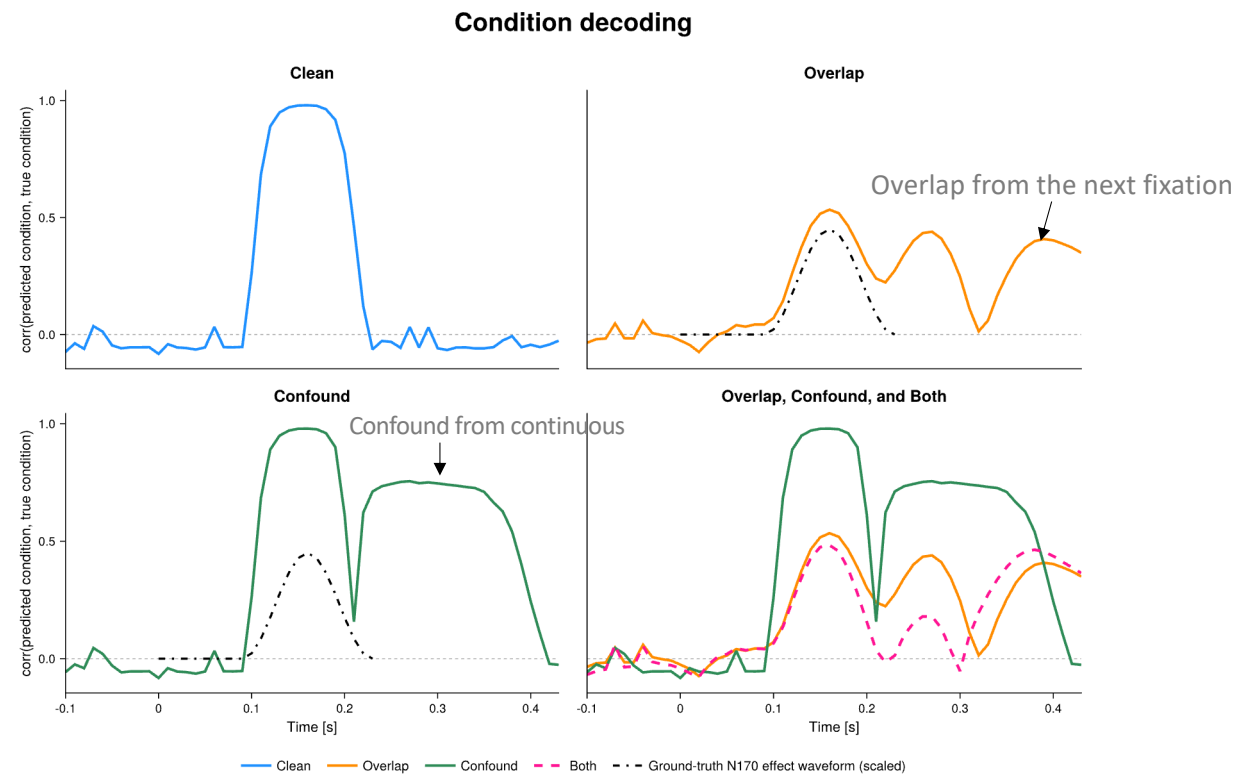
Clean: No overlap, independent predictors

Overlap: condition-biased overlap

Confound: $\text{corr}(\text{condition}, \text{continuous}) = 0.8$

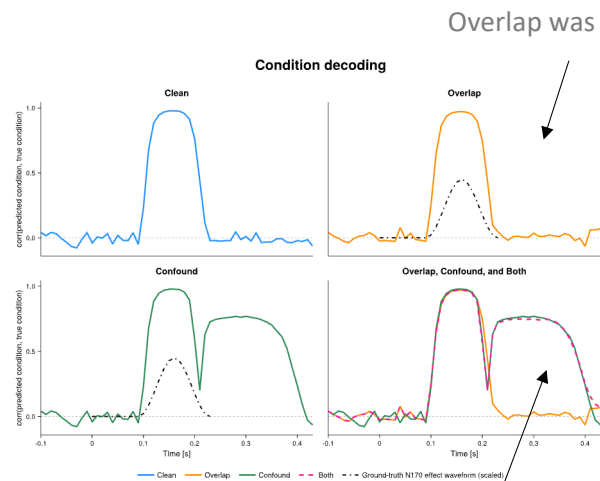
Both: overlap + correlation

Standard decoding with tuned ridge model

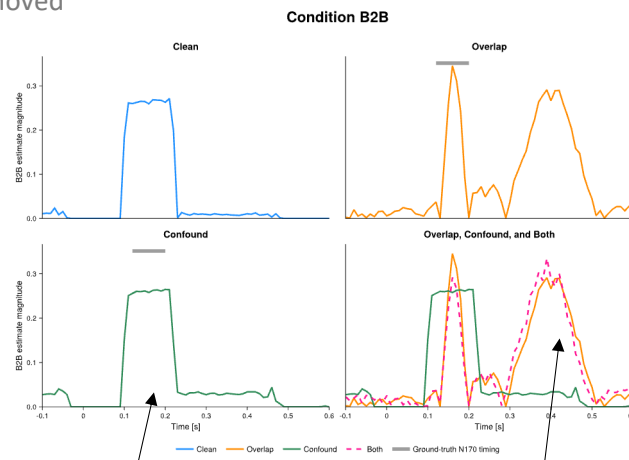


Alternative pipelines show limited recovery

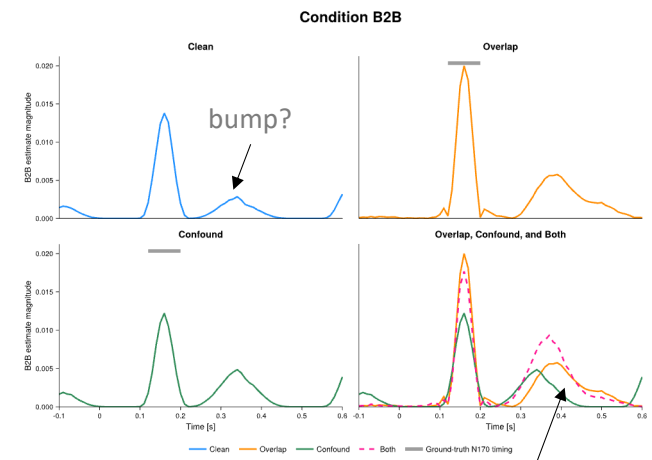
rERP decoding



Plain B2B

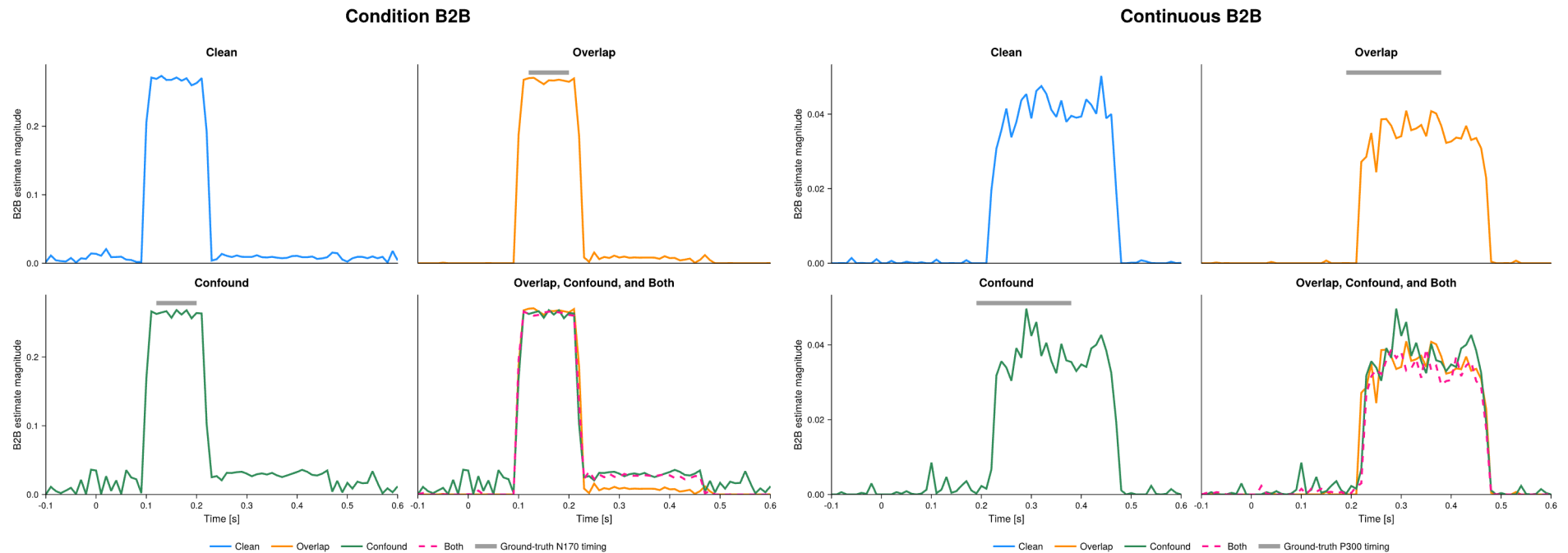


One-step B2B



Confounding bias was removed

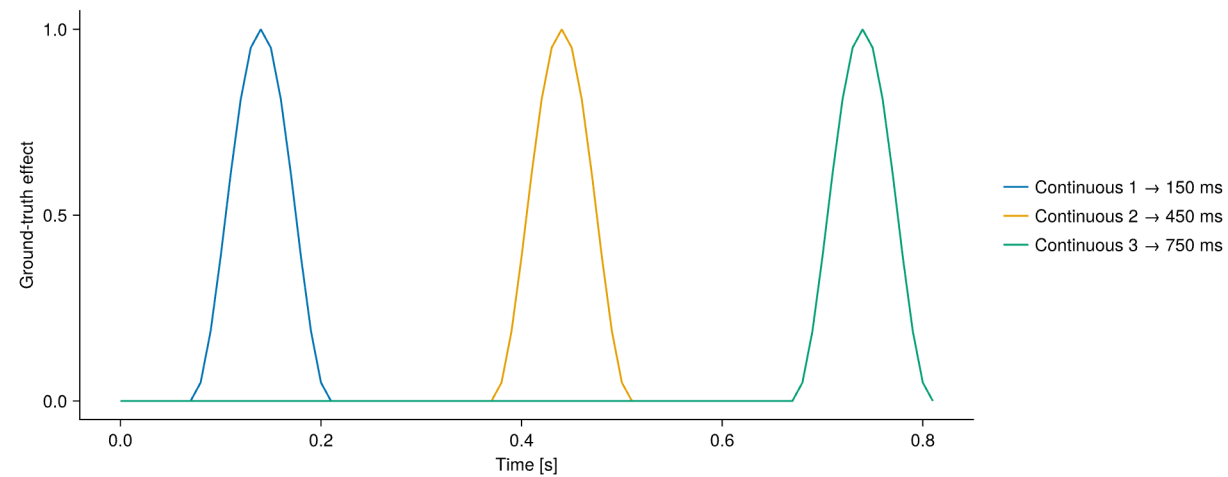
Two-step B2B recovers predictors under temporal overlap



After overlap correction, B2B can recover predictor-specific information at approximately the correct time.

Additional simulation: three correlated continuous predictors

Ground truth



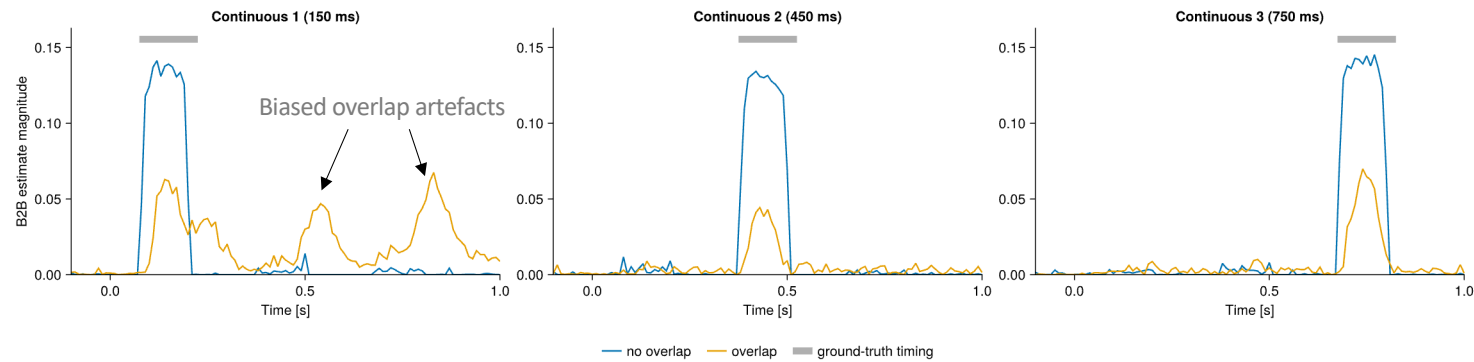
- $\rho = 0.8$ between all predictors

no_overlap: no overlap but correlated predictors

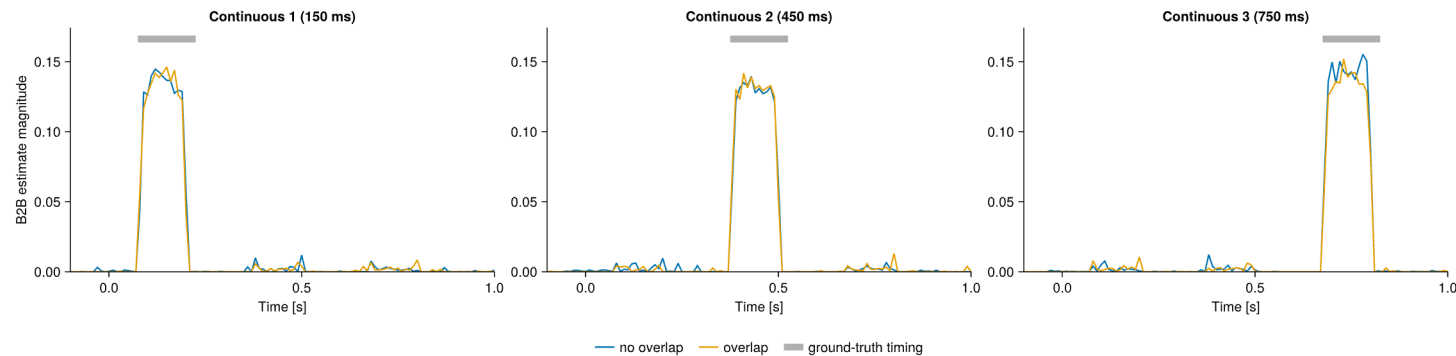
overlap: continuous 1-biased overlap and correlated predictors

Two-step B2B reduces overlap-induced distortion

Plain B2B on correlated continuous predictors



Two-step B2B on correlated continuous predictors



ROAMM dataset

ROAMM – Reading Observed at Mindless Moments

Dataset

Long continuous natural reading

EEG + ET synchronized

Predictors

Word length

Word frequency

Zipf frequency

Word surprisal

GPT-2 sentence-level context

B2B Evaluation on ROAMM

Outer loop: held-out feature importance

1. Train (X_{train}, Y_{train})

2. Full-model performance on (X_{test}, Y_{test})

3. Knock out one predictor at a time

$$\Delta R_i = R_{full} - R_{knockout(i)}$$

Inner loop: Two-step training

Fixation events & predictors

Concatenate runs within subject

rERP deconvolution

Reconstruct overlap-corrected single trials

Subject-level B2B



Conclusion

Current limitations and open questions

Simulation:

- The underlying signal remains effectively univariate instead of multivariate.

One-step B2B:

- High computational cost
- Estimation behaviour not yet fully understood

Take-home messages

- Temporal overlap and correlated predictors are both important challenges in natural-reading EEG.
- In simulation, two-step B2B showed the clearest recovery of the simulated ground-truth effects.
- Next: evaluate two-step B2B on ROAMM using held-out evaluation.

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